

UNIVERSITY OF FRONTIER TECHNOLOGY, BANGLADESH

DEPARTMENT OF INTERNET OF THINGS AND ROBOTICS ENGINEERING (IRE)

FACULTY OF CYBER PHYSICAL SYSTEM ENGINEERING

LAB REPORT 03

DART COLLECTIONS, FILE HANDLING AND CLI TO-DO APPLICATION

(Lists, Sets, Maps, File I/O, Higher-Order Functions, and CLI App)

COURSE CODE: IRE 460

COURSE TITLE: MOBILE PLATFORM FOR IOT DEVICES SESSIONAL

SUBMITTED BY:

Shahriar Alom Masud

Student ID: **2201032** | Session: 2022–2023

Department of Internet of Things and Robotics Engineering (IRE), UFTB

SUBMITTED TO:

Md. Rafiqul Islam

Lecturer

Department of Internet of Things and Robotics Engineering (IRE)

University of Frontier Technology, Bangladesh

NOVEMBER 03, 2026

Laboratory Overview

This laboratory assignment covers fundamental data structures (Lists, Sets, Maps), File I/O operations, and console-based application development in Dart for mobile and backend systems.

GitHub Repository Link:
https://github.com/Masud744/flutter-lab/tree/main/Lab_03

Problem 1: List of Names

Task: Create a list of names and print all names using list.

Source Code (src/problem1_names_list.dart):

```dart
void main() {
  List<String> names = ['Shahriar', 'Masud', 'Rafiqul', 'Tanvir', 'Sakib'];

  print('Names list: $names\n');

  print('Printing each name:');
  for (var name in names) {
    print(name);
  }
}
```

Execution Output:

```
Names list: [Shahriar, Masud, Rafiqul, Tanvir, Sakib]

Printing each name:
Shahriar
Masud
Rafiqul
Tanvir
Sakib
```

Problem 2: Set of Fruits

Task: Create a set of fruits and print all fruits using loop.

Source Code (src/problem2_fruits_set.dart):

```dart
void main() {
  Set<String> fruits = {'Apple', 'Banana', 'Mango', 'Orange', 'Strawberry'};

  print('Fruits in set:');
  for (var fruit in fruits) {
    print(fruit);
  }
}
```

Execution Output:

```
Fruits in set:
Apple
Banana
Mango
Orange
Strawberry
```

Problem 3: Expenses Total Calculator

Task: Create a program that reads list of expenses amount using user input and print total.

Source Code (`src/problem3_expenses.dart`):

```dart
import 'dart:io';

void main() {
  List<double> expenses = [];

  stdout.write('How many expenses do you want to enter? ');
  int count = int.parse(stdin.readLineSync()!);

  for (int i = 0; i < count; i++) {
    stdout.write('Enter expense ${i + 1}: ');
    double amount = double.parse(stdin.readLineSync()!);
    expenses.add(amount);
  }

  double total = 0;
  for (var e in expenses) {
    total += e;
  }

  print('\nAll expenses: $expenses');
  print('Total expense: \$${total.toStringAsFixed(2)}');
}
```

Execution Output:

```
How many expenses do you want to enter? 3
Enter expense 1: 120.50
Enter expense 2: 45.00
Enter expense 3: 84.50

All expenses: [120.5, 45.0, 84.5]
Total expense: $250.00
```

Problem 4: Days of the Week List

Task: Create an empty list of type string called `days`. Use the `add` method to add names of 7 days and print all days.

Source Code (`src/problem4_days_list.dart`):

```dart
void main() {
  List<String> days = [];

  // Adding names of 7 days
  days.add('Monday');
  days.add('Tuesday');
  days.add('Wednesday');
  days.add('Thursday');
  days.add('Friday');
  days.add('Saturday');
  days.add('Sunday');

  print('Days of the week:');
  for (var day in days) {
    print(day);
  }
}
```

Execution Output:

```
Days of the week:
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
```

Problem 5: Capitalize Words in File

Task: Read data from a file and capitalized each word and write the result in separate file. Both file should be in same format.

Source Code (`src/problem5_file_capitalize.dart`):

```dart
import 'dart:io';

String capitalizeWords(String text) {
  var words = text.split(' ');
  for (int i = 0; i < words.length; i++) {
    if (words[i].isNotEmpty) {
      words[i] = words[i][0].toUpperCase() + words[i].substring(1);
    }
  }
  return words.join(' ');
}

void main() {
  var dir = File(Platform.script.toFilePath()).parent.path;
  var inputFile = File('$dir/input.txt');
  var outputFile = File('$dir/output.txt');

  String content = inputFile.readAsStringSync();
  print('Original file content:\n$content');

  // Capitalize each word line by line
  var lines = content.split('\n');
  for (int i = 0; i < lines.length; i++) {
    lines[i] = capitalizeWords(lines[i]);
  }
  String result = lines.join('\n');

  outputFile.writeAsStringSync(result);

  print('\nCapitalized content saved to output.txt:\n$result');
}
```

Execution Output:

```
Original file content:
mobile platform for iot devices sessional
learning dart collections and file handling

Capitalized content saved to output.txt:
Mobile Platform For Iot Devices Sessional
Learning Dart Collections And File Handling
```

Problem 6: Person Map Key Update

Task: Create a map with name, address, age, country keys and store values to it. Update country name to other country and print all keys and values.

Source Code (`src/problem6_person_map.dart`):

```dart
void main() {
  Map<String, dynamic> person = {
    'name': 'Shahriar Alom Masud',
    'address': 'Dhaka, Bangladesh',
    'age': 22,
    'country': 'Bangladesh',
  };

  print('Initial map:');
  person.forEach((key, value) {
    print('$key: $value');
  });

  // Updating country
  person['country'] = 'Germany';

  print('\nAfter updating country:');
  person.forEach((key, value) {
    print('$key: $value');
  });
}
```

Execution Output:

```
Initial map:
name: Shahriar Alom Masud
address: Dhaka, Bangladesh
age: 22
country: Bangladesh

After updating country:
name: Shahriar Alom Masud
address: Dhaka, Bangladesh
age: 22
country: Germany
```

Problem 7: Filter Map Keys with `where`

Task: Create a map with name, phone keys and store some values to it. Use where to find all keys that have length 4.

Source Code (`src/problem7_phone_map.dart`):

```dart
void main() {
  Map<String, String> contacts = {
    'John': '01711223344',
    'Alex': '01822334455',
    'Masud': '01933445566',
    'Sara': '01544556677',
    'Rifat': '01655667788',
    'Liam': '01366778899',
  };

  print('All contacts:');
  contacts.forEach((name, phone) {
    print('$name: $phone');
  });

  // Find all keys with length 4
  var keysLength4 = contacts.keys.where((key) => key.length == 4);

  print('\nKeys with length 4:');
  for (var key in keysLength4) {
    print('$key: ${contacts[key]}');
  }
}
```

Execution Output:

```
All contacts:
John: 01711223344
Alex: 01822334455
Masud: 01933445566
Sara: 01544556677
Rifat: 01655667788
Liam: 01366778899

Keys with length 4:
John: 01711223344
Alex: 01822334455
Sara: 01544556677
Liam: 01366778899
```

Problem 8: Simple To-Do Application

Task: Create a simple to-do application that allows user to add, remove, and view their task.

Source Code (`src/problem8_todo_app.dart`):

```dart
import 'dart:io';

void main() {
  List<String> tasks = [];

  while (true) {
    print('\n--- To-Do App ---');
    print('1. View Tasks');
    print('2. Add Task');
    print('3. Remove Task');
    print('4. Exit');
    stdout.write('Enter choice (1-4): ');

    String? choice = stdin.readLineSync();

    if (choice == '1') {
      if (tasks.isEmpty) {
        print('No tasks available.');
      } else {
        print('\nYour Tasks:');
        for (int i = 0; i < tasks.length; i++) {
          print('${i + 1}. ${tasks[i]}');
        }
      }
    } else if (choice == '2') {
      stdout.write('Enter task: ');
      String? task = stdin.readLineSync();
      if (task != null && task.trim().isNotEmpty) {
        tasks.add(task.trim());
        print('Task added.');
      }
    } else if (choice == '3') {
      if (tasks.isEmpty) {
        print('No tasks to remove.');
      } else {
        stdout.write('Enter task number to remove: ');
        int? index = int.tryParse(stdin.readLineSync() ?? '');
        if (index != null && index >= 1 && index <= tasks.length) {
          String removed = tasks.removeAt(index - 1);
          print('Removed: $removed');
        } else {
          print('Invalid task number.');
        }
      }
    } else if (choice == '4') {
      print('Goodbye!');
      break;
    } else {
      print('Invalid option. Please choose 1-4.');
    }
  }
}
```

Problem 8 Execution Output

Below is the console session demonstration showing adding a task, viewing the active list, removing a completed task, and exiting:

```
--- To-Do App ---
1. View Tasks
2. Add Task
3. Remove Task
4. Exit
Choose option (1-4): 2
Enter task: Finish lab report
Task added.

--- To-Do App ---
1. View Tasks
2. Add Task
3. Remove Task
4. Exit
Choose option (1-4): 1

Your Tasks:
1. Finish lab report

--- To-Do App ---
1. View Tasks
2. Add Task
3. Remove Task
4. Exit
Choose option (1-4): 3
Enter task number to remove: 1
Removed: Finish lab report

--- To-Do App ---
1. View Tasks
2. Add Task
3. Remove Task
4. Exit
Choose option (1-4): 1
No tasks available.

--- To-Do App ---
1. View Tasks
2. Add Task
3. Remove Task
4. Exit
Choose option (1-4): 4
Goodbye!
```

Conclusion

In this laboratory session, all eight exercises covering Dart collections (Lists, Sets, Maps), predicate-based filtering using the `where` method, synchronous file reading and writing, and console menu-driven application logic were implemented and verified. The code demonstrates clean and efficient Dart programming suitable for mobile application backends.